INCH-POUND

MIL-S-44384
AMENDMENT 1
8 May 1990

MILITARY SPECIFICATION

SUIT, CONTAMINATION AVOIDANCE, LIQUID PROTECTION (SCALP)

This amendment forms a part of Military Specification MIL-S-44384, dated 16 November 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

3.3.2, line 2, after "pigmented": Insert ",with a minimum thickness of 0.0015 inch,".

Table I, in characteristic column, lines 7 and 8: Delete "(lbs/in)" and substitute "(lbs)" in two places.

PAGES 4 and 5

Table I: Delete test method column entirely.

PAGES 5 and 6

Table I: Delete footnotes 1/ through 9/ in entirety.

PAGE 15

Table IV, footnote 1/: Delete "center back" and substitute "neckline (top of the shoulder)".

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:52Z
Check the source to verify that this is the current version before use.

4.4.1.1: Delete table and substitute:

"

Component	Characteristic	Requirement	Test Method
Base cloth	Weight, oz/sq.yd	3.3.1 and Table I	ASTM D 3776 1/
	Breaking load, cut strip method		
	Machine dir.(lbs/in)	3.3.1 and Table I	ASTM D 1117 2/
	Transverse dir. (lbs/in)	3.3.1 and Table I	ASTM D 1117 2/
	Tongue tear strength		
	Machine dir. (lbs)	3.3.1 and Table I	ASTM D 2261 3/
	Transverse dir. (lbs)	3.3.1 and Table I	ASTM D 2261 3/
	Water resistance hydrostatic head, in.	3.3.1 and Table I	AATCC 127
	Electrical resistivity Log R	3.3.1 and Table I	AATCC 76 4/
Coated compound		3.3.2	----
Latex		3.3.7	----
Cord locks		3.3.8	----

1/ Square yard.

2/ Cut strip; CRE tensile test, crosshead speed 2 in/min, guage length 5 inches.

3/ Run at crosshead speed of 5 in/min and 1 inch gauge length.

4/ Converted to Log R.".

Add the following new paragraph:

"4.4.1.2 Coated cloth testing. Every roll of coated cloth shall have a sample of 3 continuous yards full width removed for testing in accordance with Table IVa. Any test failure shall be cause for rejection of the roll from which the failed sample was taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:52Z
Check the source to verify that this is the current version before use.

TABLE IVa. Coated cloth testing

Characteristic	Requirement	Test Method
Thickness, inch	Table I	ASTM D 2103
Weight, oz/sq. yd	Table I	ASTM D 3776
Min tensile strength (lb/in) 1/		
Initial:		
Machine dir.	Table I	ASTM D 1682
Transverse dir.	Table I	ASTM D 1682
After oven aging:		
Machine dir.	Table I	FED-STD-191-5850 5/
Transverse dir.	Table I	FED-STD-191-5850 5/
Tear strength, lb		
Machine and transverse dir.	Table I	ASTM D 2261
Bursting strength, psi	Table I	ASTM D 3786
Resistance to leakage:		
Initial	Table I	FED-STD-191-5516 2/
After cold crack	Table I	FED-STD-191-5874 3/
Resistance to blocking rating	Table I	FED-STD-191-5872
Resistance to permeation toxicological agent (min), minutes 4/:		
Mustard (HD)	Table I	CRDC-SP-84010 & 4.5.4
Thickened GD	Table I	CRDC-SP-84010 & 4.5.5
VX	Table I	CRDC-SP-84010 & 4.5.6

1/ One-inch wide cut strip.

2/ The polyethylene coated side shall contact the water. The hydrostatic pressure shall be 50 cm for 5 minutes. The appearance of one water drop at any place within the 4-1/2 inch diameter test area shall be reported as a test failure.

3/ Conditioned for 30 minutes at -40° prior to testing.

4/ Chemical Research Development and Engineering Center Test.

5/ Temperature at 180° for 7 days.".

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:52Z
Check the source to verify that this is the current version before use.

Add the following new paragraph and table:

"4.4.1.3 Coated cloth toxicological agents resistance testing. Every roll of coated cloth shall have a sample of 3 continuous yards full width removed and forwarded to the U.S. Army Armament Munitions and Chemical Command (Edgewood Area) for testing in accordance with table IVb. Any test failure shall be cause for rejection of the roll from which the failed sample was taken. All test reports shall contain the individual values used in expressing the final results (see 6.8).

TABLE IVb. Coated cloth toxicological agents resistance tests

Characteristic	Requirement paragraph	Test Method
Resistance to permeation by toxicological agent, min:		
Mustard (HD)	Table I	4.5.4
Thickened GD	Table I	4.5.5
VX	Table I	4.5.6

"

PAGE 21

Add the following new paragraphs:

"4.5.4 Resistance to mustard (HD). Resistance to permeation by mustard (HD) shall be determined as specified in CRDC-SP-84010, "Explusion Test Procedure", except that the chemical agent challenge shall be 10 grams per square meter applied in ten 1.0 milligram droplets on the coated side of the cloth. In addition, the weight shall be applied to the sample for the entire test duration (1 hour). Test specimens shall be obtained by quartering a 3 yard piece of the coated cloth and selecting two 2 inch diameter circular specimens, at random, from each quarter. All specimens shall meet the mustard toxicological requirement listed in table I. In the event that seven of the tested specimens meet the requirements but one does not, four additional test specimens from each quarter shall be tested. If all sixteen specimens meet the requirements, the material shall be considered acceptable.

4.5.5 Resistance to thickened GD. Resistance to permeation by thickened GD shall be determined as specified in CRDC-SP-84010, "Explusion Test Procedure", except that the chemical agent challenge shall be 10 grams per square meter applied in one 10 milligram droplet on the coated side of the cloth. In addition, the weight shall be applied to the sample for the entire test duration (1 hour). Test specimens shall be obtained by quartering a 3 yard piece of the coated cloth and selecting two 2 inch diameter circular specimens, at random,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:52Z
Check the source to verify that this is the current version before use.

from each quarter. No test specimen shall be taken within 2 inches of the selvage edge. All specimens shall meet the thickened GD toxicological requirements but one does not, four additional test specimens from each quarter shall be tested. If all sixteen specimens meet the requirements, the material shall be considered acceptable.

4.5.6 Resistance to VX. Resistance to permeation by VX shall be determined as specified in CRDC-SP-84010, "Explusion Test Procedure", except that the chemical agent challenge shall be 10 grams per square meter applied in ten 1.0 milligram droplets on the coated side of the cloth. In addition, the wieght shall be applied to the sample for the entire test duration (1 hour). Test specimens shall be obtained by quartering a 3 yard piece of the coated cloth and selecting two 2 inch diameter circular specimens, at random, from each quarter. No test specimens shall be taken within 2 inches of the selvage edge. All specimens shall meet the VX toxicological requirements listed in table I. In the event that seven of the tested specimens meet the requirements but one does not, four additional test specimens from each quarter shall be tested. If all sixteen specimens meet the requirements, the material shall be considered acceptable.".

PAGE 25

Add the following new paragraph:

"6.8 Toxicological tests. Because of the extreme dangers involved in handling toxicological warfare agents, and the special equipment required in connection with their use, all document tests involving toxicological agents will be performed by the U.S. Army Armement Munitions and Chemical Command (Edgewood Area), Chemical Testing Branch, Building E5100, Aberdeen Proving Ground, MD 21010-5423 or an authorized independent testing laboratory approved by the U.S. Army Armament Munitions and Chemical Command.".

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - EA, MD
Air Force - 82, 11
DLA - CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

(Project 8415-0771)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:52Z
Check the source to verify that this is the current version before use.